

1

Archive and standardize

coordinates · time · units · identifiers

2

Reconstruct static network

buses · lines · transformers · parameters

3

Map assets and align time

generation · load · storage · exchange

4

Assemble and solve cases

modal P/Q · controls · AC power flows

5

Store and validate

indexed database · provenance · quality checks

Cross-cutting audit:

source key · evidence grade · transformation rule · coverage · conservation · convergence